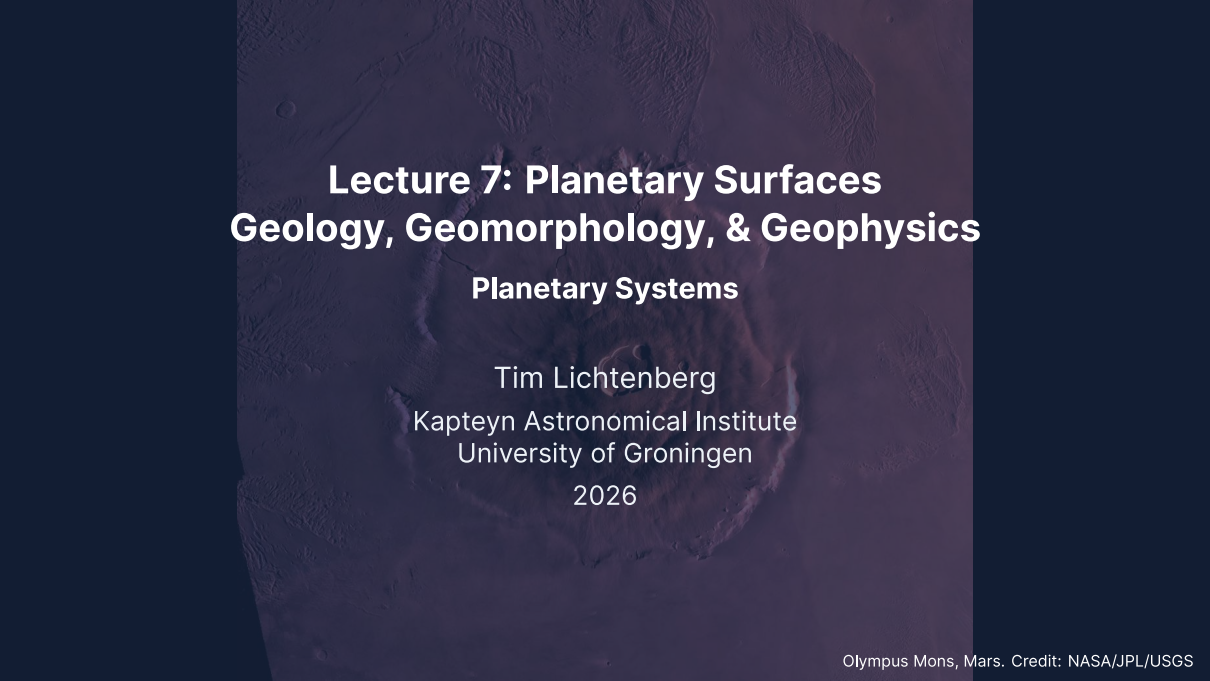


The background of the slide is a high-resolution, false-color image of the Martian surface, specifically focusing on the Olympus Mons volcano. The volcano's massive, layered rim and central caldera are visible, surrounded by a vast, desolate landscape with various smaller craters and ridges. The image is tinted with a dark, reddish-purple hue, giving it a dramatic and scientific appearance.

Lecture 7: Planetary Surfaces Geology, Geomorphology, & Geophysics

Planetary Systems

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University of Groningen

2026

Learning objectives

By the end of this lecture, you will be able to:

- Distinguish endogenic and exogenic surface processes and the three stages of crater formation
- Derive the crater scaling law $D \sim (E/\rho g)^{1/4}$ and use crater counting for surface ages
- Compare volcanic, tectonic, and erosional styles across the solar system

Recap: Lectures 5 & 6

- Hydrostatic equilibrium and the scale height set an atmosphere's structure
- Greenhouse warming: $T_s > T_{\text{eff}}$; Clausius-Clapeyron sets where clouds form
- The carbonate-silicate cycle is Earth's long-term thermostat

Today: What shapes the *surfaces* of planets and moons?



1. Surface processes

Valles Marineris hemisphere of Mars, Viking Orbiter mosaic. Credit: NASA/JPL/USGS

Endogenic vs. exogenic processes

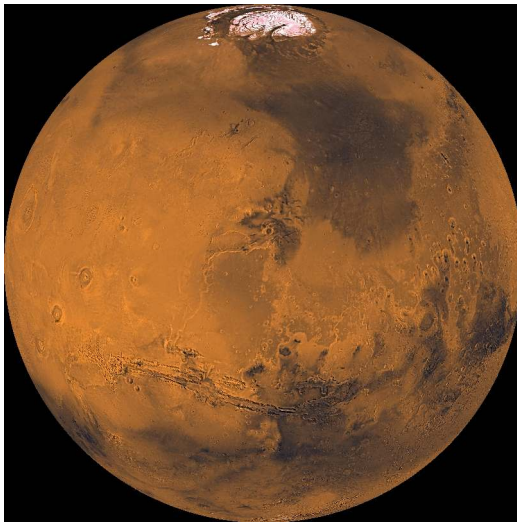
A surface is a geological record: processes that build landforms compete with processes that destroy them.

- **Endogenic** (internal heat): volcanism, tectonics (faulting, folding)
- **Exogenic** (external forces): impact cratering, erosion by wind, water and ice

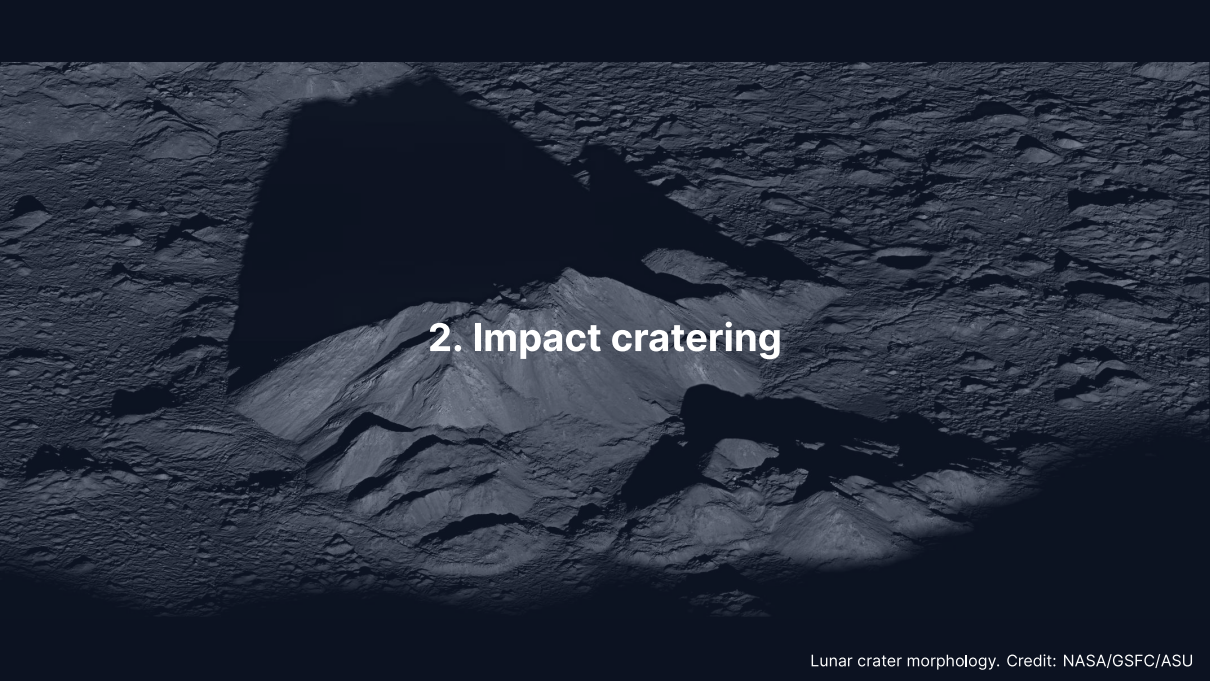
Hot interior: endogenic processes win. Dead interior: the surface keeps its ancient craters.

Dominant surface process by body

Body	Dominant process	Key evidence
Moon	Impact cratering	Saturated highland craters, >4 Gyr
Earth	Plate tectonics + erosion	Ocean floors <200 Myr; rapid weathering
Mars	Volcanism + impacts	Olympus Mons; cratered southern highlands
Venus	Volcanism	Volcanic plains cover ~80% of surface
Io	Tidal volcanism	~300–400 active volcanoes; age <1 Myr
Europa	Ice tectonics	Lineae, chaos terrain, very few craters



Mars shows **all** four surface processes on one disk: volcanism at Tharsis, rifting at Valles Marineris, cratering in the southern highlands, polar erosion. Viking Orbiter mosaic. Credit: NASA/JPL/USGS.

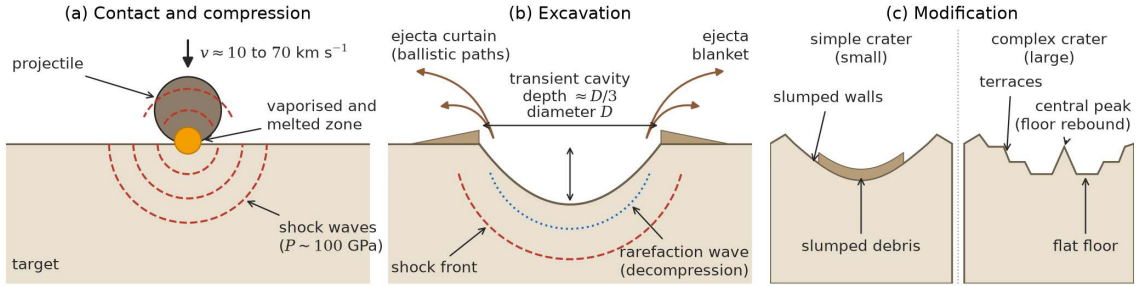
A black and white photograph of a lunar crater. The crater floor is covered in a dense field of smaller craters and rocks. A large, dark shadow is cast across the upper left portion of the image. In the center, a prominent, conical central peak rises from the floor. The text "2. Impact cratering" is overlaid in white on the central peak.

2. Impact cratering

The most universal geological process

- **Every** solid body in the solar system bears impact craters
- Without atmosphere or geology (Moon, Mercury), craters survive billions of years
- Impact velocities $10\text{--}70\text{ km s}^{-1}$, set by orbital mechanics (Lecture 2)

A 1 km rocky asteroid at 20 km s^{-1} carries $E_k = \frac{1}{2}mv^2 \approx 3 \times 10^{20}\text{ J}$, about 1000 times the largest nuclear weapon, released in under a second.



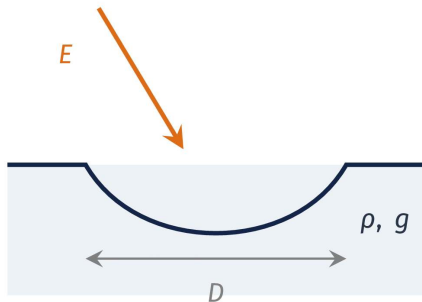
Three stages of crater formation: contact and compression, excavation of a transient cavity, modification into a simple or complex crater. Course figure.

Contact and compression (shock near 100 GPa, over in under 1 s), excavation of a transient cavity (depth:diameter about 1:3, ejecta blanket at the rim), modification: walls slump into a **simple crater** or the floor rebounds into a **central peak**.

3. Blackboard derivation: the crater scaling law

Caloris basin, Mercury. Credit: NASA/Johns Hopkins APL/Carnegie Institution of Washington

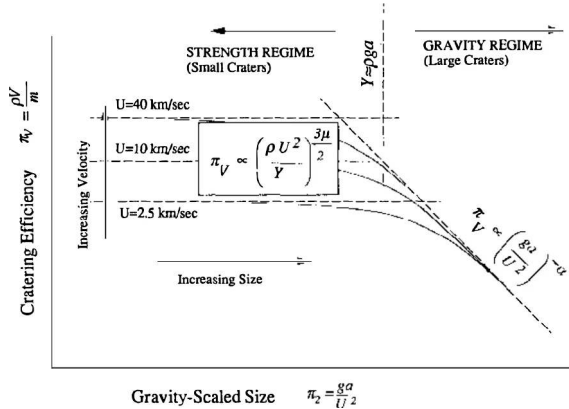
Goal and strategy



Goal: how does crater diameter D depend on impact energy E , target density ρ and gravity g ?

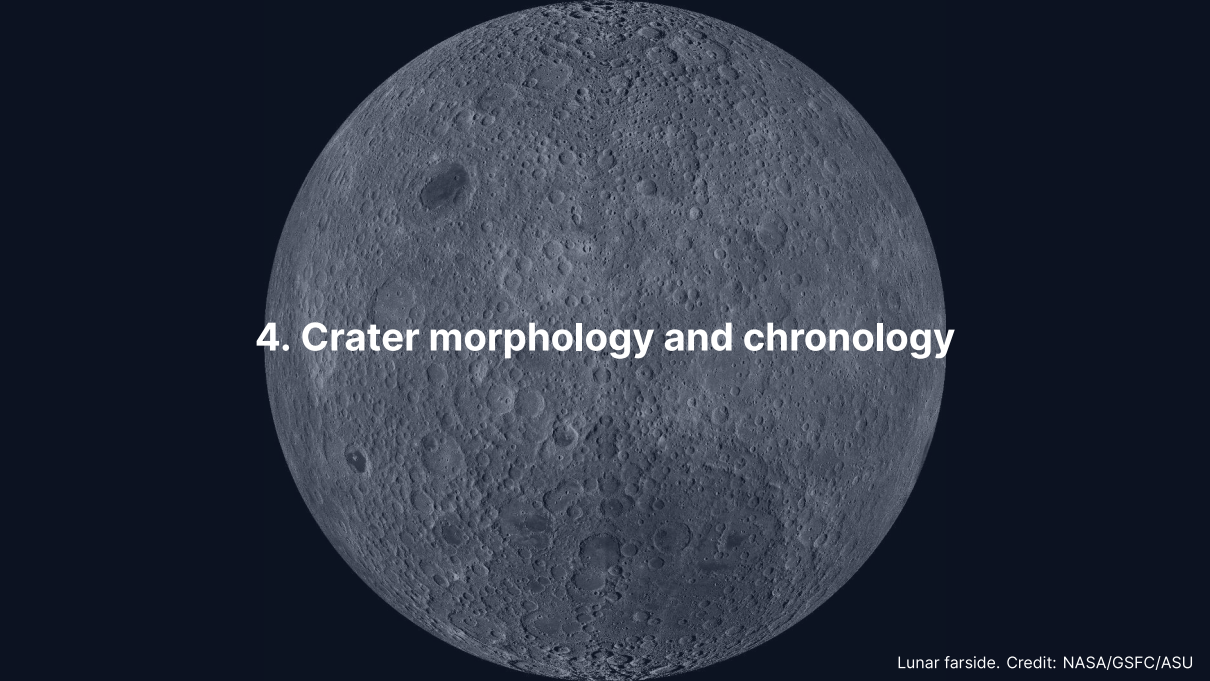
- **Gravity regime** (craters $\gtrsim 100$ m):
 $D = C E^a \rho^b g^c$
- Dimensions: $[E] = ML^2T^{-2}$,
 $[\rho] = ML^{-3}$, $[g] = LT^{-2}$
- Three equations for a, b, c ; then numbers

To the board.

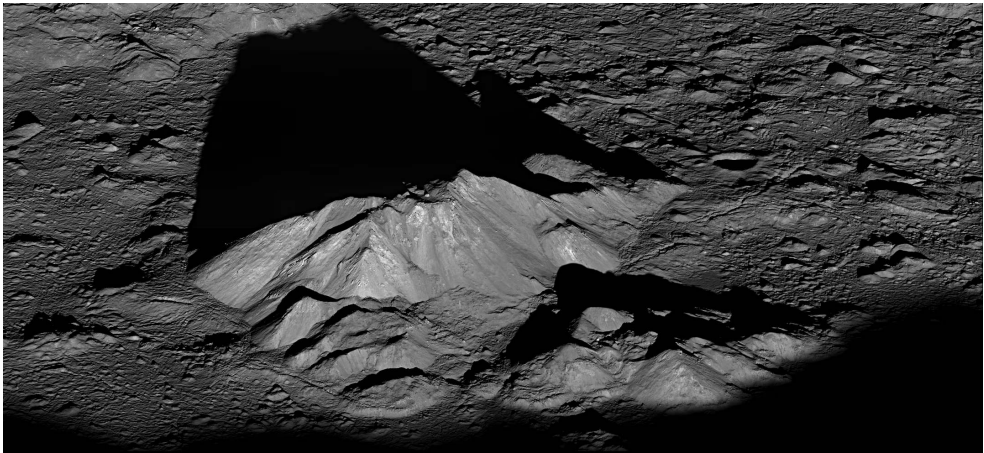


Crater size against impactor size in the strength and gravity regimes, curves for $U = 2.5, 10$ and 40 km s^{-1} . Holsapple (1993), Fig. 3.

$D \sim (E/\rho g)^{1/4}$ holds in the gravity regime, above about 100 m on the Moon: doubling E grows D by only $2^{1/4} \approx 1.19$, and at fixed v and ρ a 1 km impactor makes a 16.5 km crater ($D \propto L^{3/4}$).

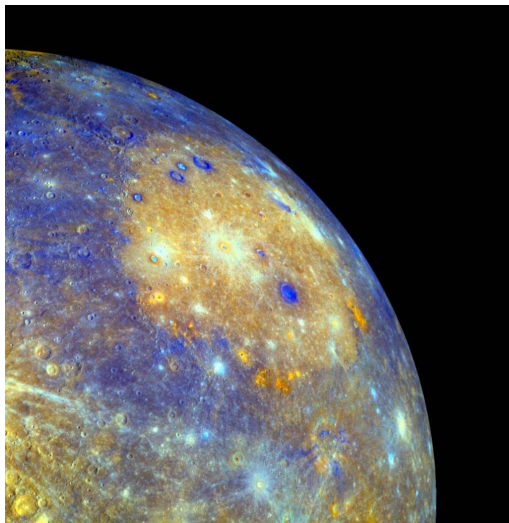


4. Crater morphology and chronology



Simple, complex and multi-ring craters on the Moon. Credit: NASA/GSFC/ASU.

Simple bowls (depth:diameter about 1:5) below $D_t \approx 15$ km on the Moon; **complex** craters with central peak, terraces and flat floor up to about 300 km; **multi-ring basins** beyond: Orientale 930, Caloris 1550, Hellas 2300 km.



Caloris basin, Mercury, about 1550 km across: concentric rings, and volcanic plains (orange) that filled it after the impact. Credit: NASA/Johns Hopkins APL/Carnegie Institution of Washington.

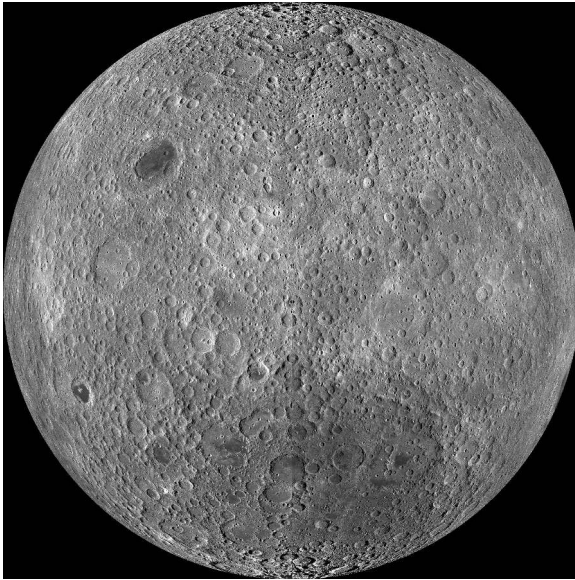
Transition diameter

The simple-to-complex transition scales with surface gravity:

$$D_t \approx D_{t,\text{Moon}} \cdot \frac{g_{\text{Moon}}}{g}$$

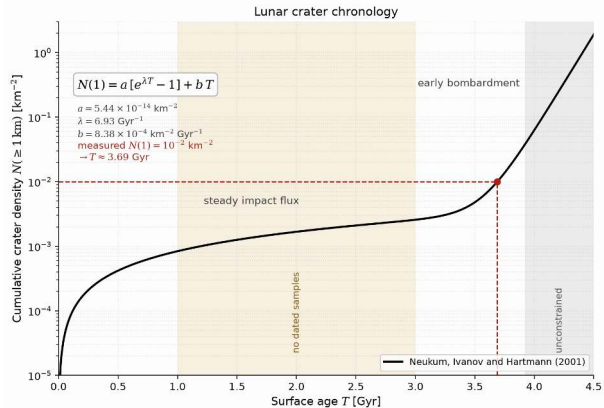
Body	$g \text{ (m s}^{-2}\text{)}$	$D_t \text{ (km)}$
Moon	1.62	15
Mars	3.71	~6.5
Earth	9.81	~2.5
Mercury	3.70	~6.6

On Earth, virtually all craters larger than a few km are complex.



Crater counting on the lunar farside: older surfaces carry more craters. Credit: NASA/GSFC/ASU; method after Neukum et al. (2001).

$N(> D) = a D^{-b}$ with $b \approx 2-3$, a straight line on log-log axes: a higher line is an older surface, so counts give **relative** ages until **saturation**, and Apollo and Luna samples supply the **absolute** calibration.



The lunar cratering chronology $N(1) = a[e^{\lambda T} - 1] + bT$. Course figure, after Neukum, Ivanov and Hartmann (2001).

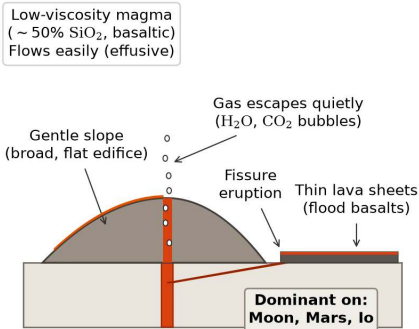
The linear term is the steady flux of the last 3 Gyr, the exponential term the early bombardment before 3.6 Gyr; $N(1) = 10^{-2} \text{ km}^{-2}$ reads as 3.7 Gyr, and the shaded bands hold no dated sample: interpolation, not calibration.



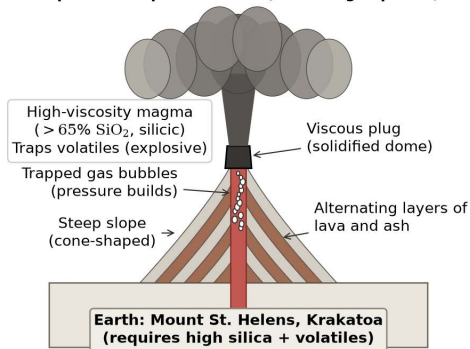
5. Volcanism

Olympus Mons, Mars. Credit: NASA/JPL/USGS

(a) Effusive volcanism (schematic)

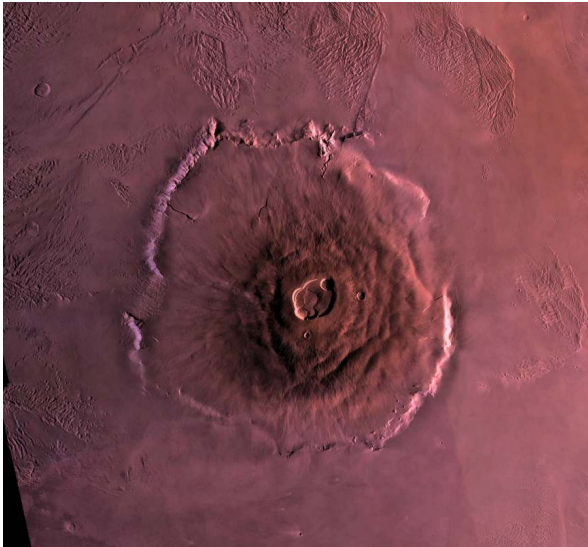


(b) Explosive volcanism (schematic) Explosive eruption column (ash and gas plume)



Effusive against explosive volcanism: silica content sets the magma viscosity, and viscosity sets whether volatiles escape quietly or build pressure. Course figure.

Basaltic magma (about $50\% \text{ SiO}_2$) flows and lets volatiles escape: **effusive** shields and flood basalts on the Moon, Mars and Io; silicic magma ($>65\% \text{ SiO}_2$) traps them until failure: **explosive** stratovolcanoes and ash, mainly on Earth.



Olympus Mons, Mars: a shield volcano about 600 km wide and 21 km high, with nested summit calderas. Viking Orbiter mosaic. Credit: NASA/JPL/USGS.



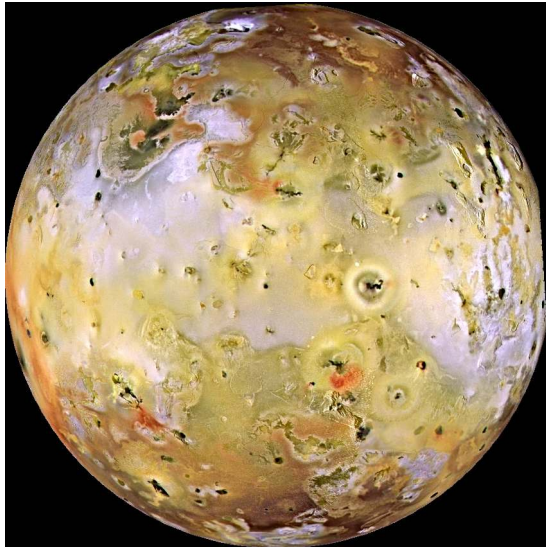
Olympus Mons (21.2 km), Mauna Kea (10.2 km from the sea floor) and Everest (8.8 km) on one height scale, widths schematic. Course figure.

No plate motion: one plume feeds one edifice for billions of years, and the lower gravity, $g \approx 0.38 g_{\oplus}$, lets the crust hold a pile about $2.6\times$ taller.

Volcanism across the solar system

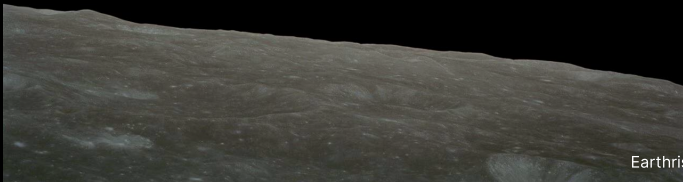
Body	Style	Driving mechanism	Examples
Earth	Effusive + explosive	Internal heat + recycling	Hawaii, mid-ocean ridges
Moon	Flood basalts (extinct)	Residual heat (3.9–3.1 Ga)	Maria
Mars	Shield volcanoes	Mantle plumes, no plates	Olympus Mons, Tharsis
Venus	Lava plains + shields	Internal heat, stagnant lid	Maat Mons
Io	Ultramafic flows	Tidal heating	Loki Patera, Pele

Venus returns in Lecture 9, Io in Lecture 11.



Io from Galileo SSI: active sulfur and silicate volcanism powered by tidal dissipation in the Laplace resonance. Credit: NASA/JPL-Caltech/University of Arizona.

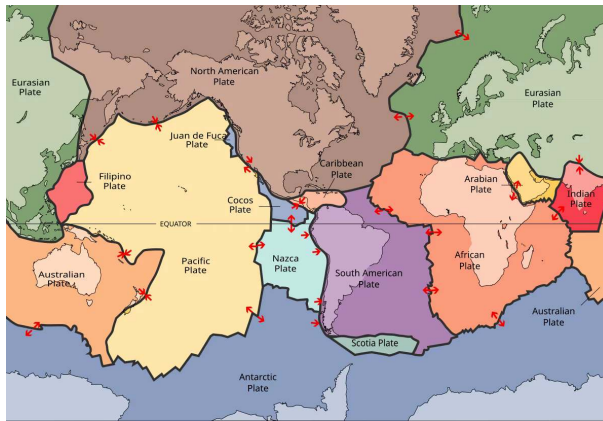
Break



An aerial photograph of the Carrizo Plain in California, showing the San Andreas Fault as a prominent, winding linear depression. The landscape is arid and flat, with some small, dark, irregular features that might be water or wet mud. The fault line runs diagonally from the upper right towards the lower left, with a slight curve. The surrounding terrain is a mix of light brown and tan colors, with some darker patches. The overall scene is a vast, open landscape with a clear, straight line of geological significance.

6. Tectonics

The San Andreas Fault across the Carrizo Plain, California. Credit: Ikluft, Wikimedia Commons, CC BY-SA 4.0



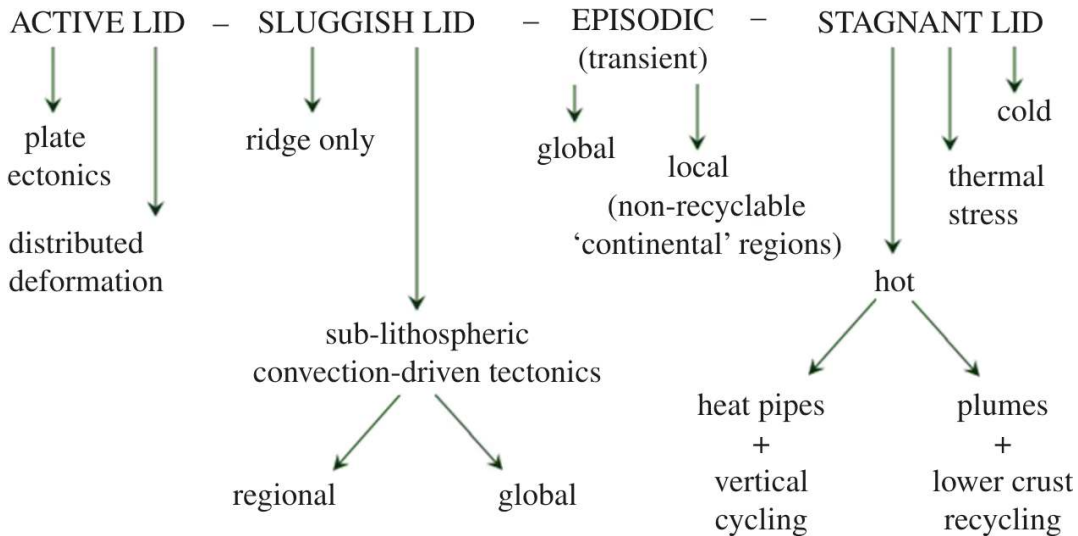
The tectonic plates of Earth and their boundaries. Credit: Wikimedia, CC BY-SA 3.0.

Earth is the **only** body with active plate tectonics: about 15 rigid plates at $1\text{--}10\text{ cm yr}^{-1}$, **divergent** ridges that make crust, **convergent** subduction that recycles it, **transform** faults that slide; subduction closes the **carbonate-silicate cycle**.

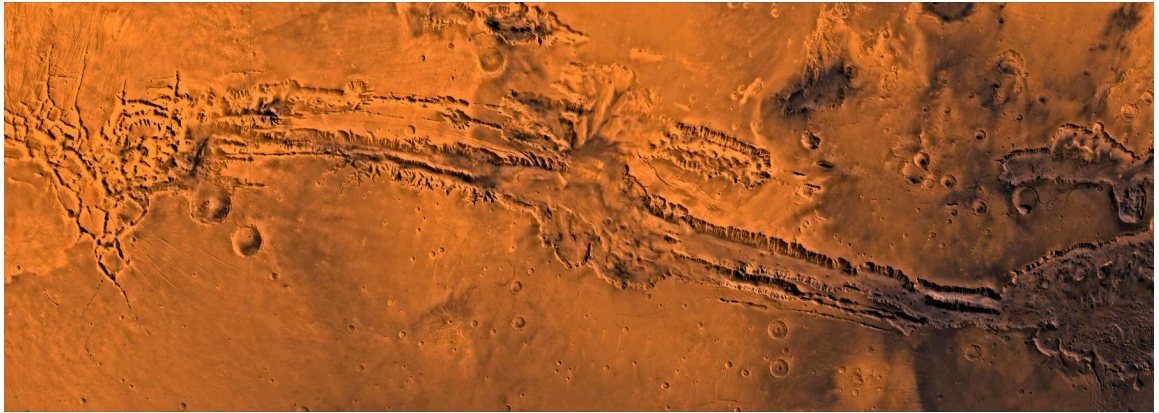
Stagnant-lid convection

- Every other terrestrial body: one rigid lithospheric lid over a convecting mantle
- Heat leaves by conduction and occasional volcanism; the thick lid keeps the mantle warm
- Melt reaches the surface only through rare plumes (Tharsis), and volcanism fades as the lid grows

Stagnant lid is the default for temperature-dependent viscosity. Breaking the lid, as Earth does, needs water weakening and self-sustaining damage.

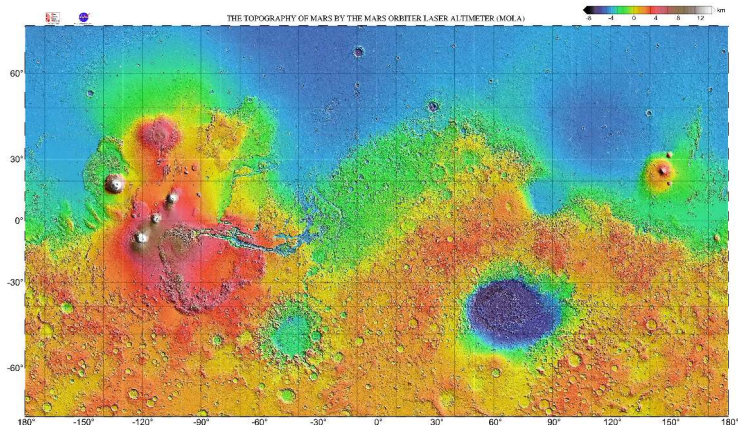


Tectonic modes of terrestrial planets beyond the mobile-stagnant binary: active lid, sluggish lid, episodic regimes, and stagnant lid (cold or hot). Reproduced from Lenardic (2018), Fig. 2.



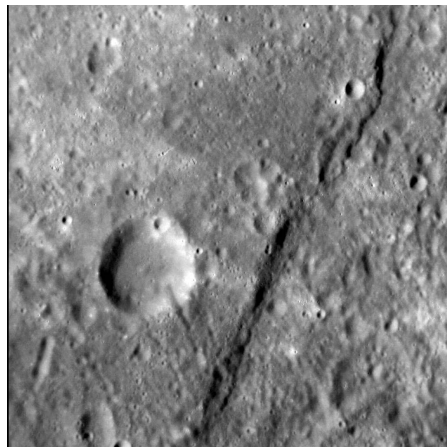
Valles Marineris in a Viking Orbiter mosaic, about 4000 km long and up to 7 km deep: no plate boundaries, yet the largest rift in the solar system, opened by the Tharsis volcanic load.

Credit: NASA/JPL/USGS.



MOLA topography of Mars: the hemispheric dichotomy. Credit: NASA/JPL/GSFC/MOLA Science Team.

About 6 km between the southern highlands and the northern lowlands; Tharsis carries four giant volcanoes, Hellas reaches -8.2 km, Valles Marineris sits at the equator.



A lobate scarp on Mercury, a thrust fault that cuts across craters, MESSENGER. Credit: NASA/Johns Hopkins APL/Carnegie Institution of Washington.

The contraction is younger than the craters it cuts: the cooling iron core shrank Mercury's radius by up to about 7 km.

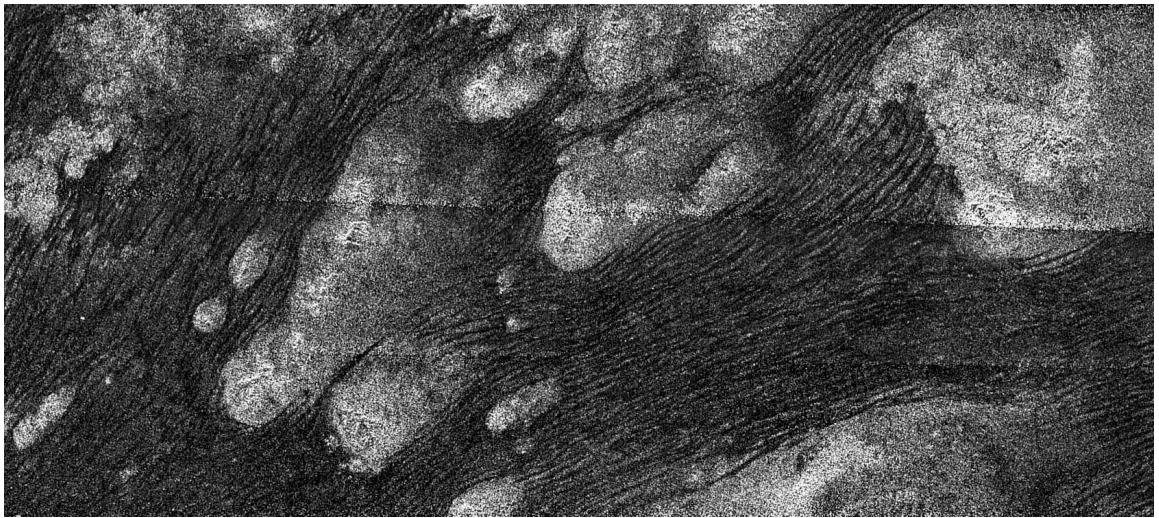


7. Erosion and weathering

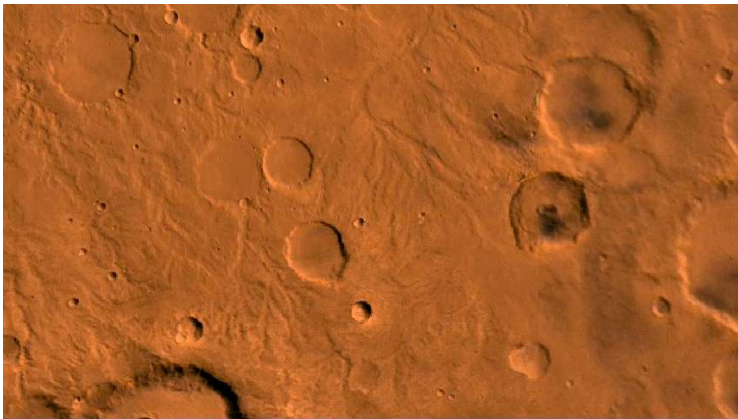
Barchan dunes in Nili Patera, Mars, imaged by HiRISE. Credit: NASA/JPL-Caltech/University of Arizona



Barchan dunes in Nili Patera, Mars, HiRISE: repeat images show whole dunes advancing at Earth-like sand fluxes under an atmosphere 170 times thinner (Bridges et al. 2012). Credit: NASA/JPL-Caltech/University of Arizona.



Longitudinal dunes in Titan's Shangri-La region, Cassini SAR: organic sand, water-ice bedrock, the same aeolian physics. Credit: NASA/JPL-Caltech/ASI.

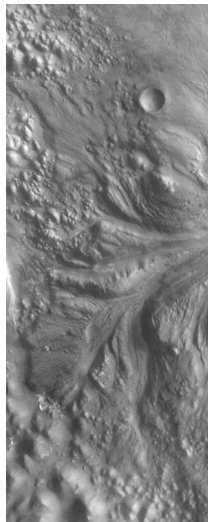


Dendritic valley networks in the Noachian highlands of Mars, Viking Orbiter. Credit: NASA/JPL/USGS.

Mars valley networks (>3.7 Ga) record sustained liquid water, **outflow channels** hundreds of km long record catastrophic groundwater release, and on **Titan** methane rivers rounded the ice pebbles Huygens landed on in 2005.

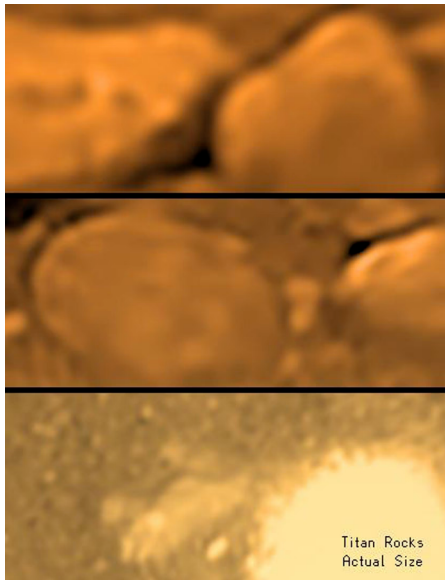


The Colorado River in the Grand Canyon has cut about 1.6 km through flat-lying Paleozoic sediments in 5 to 6 Myr, the type example of sustained fluvial erosion; resistant limestone stands as cliffs, weaker shale erodes to slopes. Credit: Alex Demas/USGS, public domain.



An outflow channel through Aram Chaos: braided islands and scoured troughs record catastrophic floods, in contrast to the dendritic valley networks that record precipitation.

Credit: NASA/JPL-Caltech/MSSS.



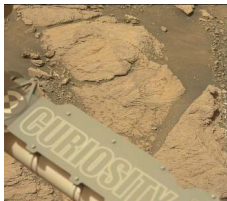
The surface of Titan from ESA's Huygens probe, 14 January 2005, in three enlarged crops. Credit: ESA/NASA/JPL-Caltech/University of Arizona, public domain.

The only image from the surface of an outer-solar-system body: decimetre blocks of water ice, rounded by methane fluvial transport, on a damp plain of organic sediment in a dried-out riverbed; liquid methane shapes Titan today.

Glacial and chemical weathering



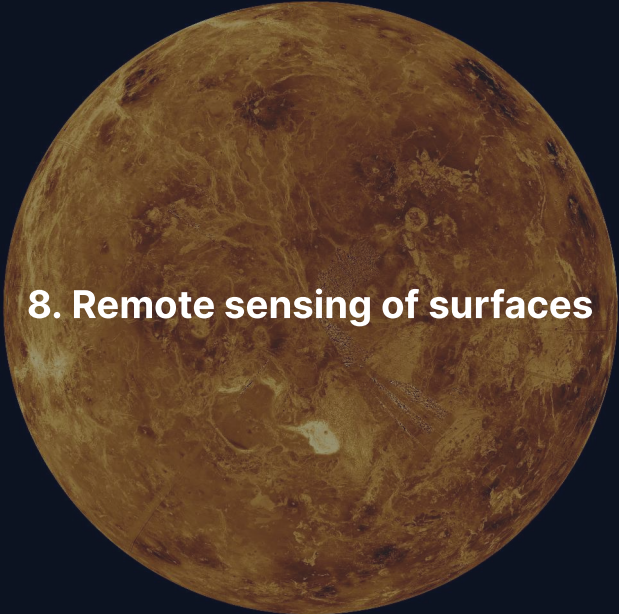
Fedchenko Glacier, Pamir. Credit: NASA/ISS



Clay-bearing unit, Glen Torridon, Gale crater. Credit:

NASA/JPL-Caltech/MSSS

- **Glacial:** ice flows under its own weight; Earth's ice sheets covered about 30% of land at the Last Glacial Maximum, Mars has polar caps and buried mid-latitude ice
- **Chemical:** silicate weathering is Earth's carbon sink; on Mars, clays, sulfates and carbonates seen from orbit record past liquid water



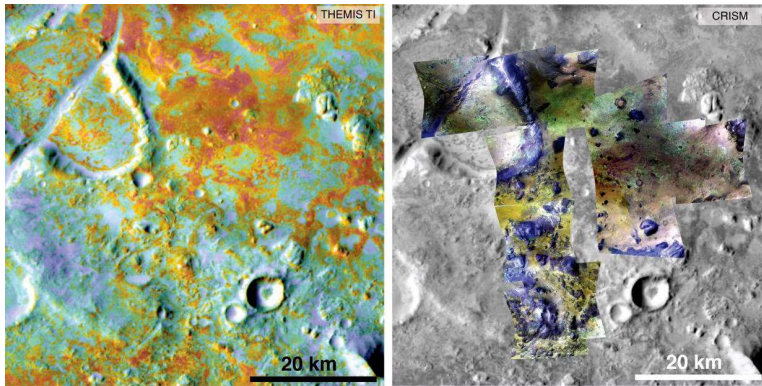
8. Remote sensing of surfaces

Venus in Magellan radar, the surface seen through an opaque atmosphere. Credit: NASA/JPL



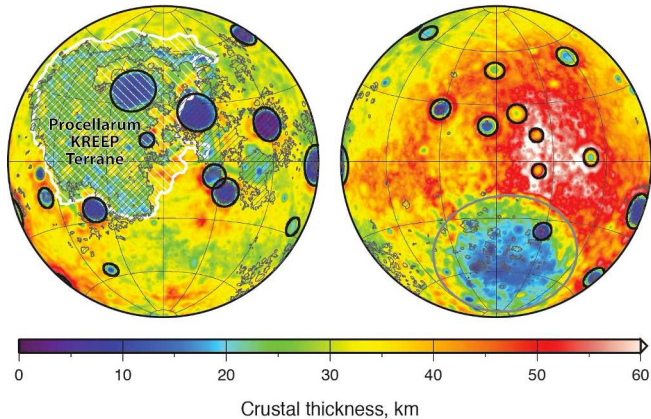
Venus from the Magellan radar mission (1990 to 1994), coloured from Venera lander measurements. Credit: NASA/JPL, public domain.

Radar imaging sees through clouds and haze: 98% of Venus mapped at about 100 m resolution, more than 1600 volcanic centres, lava plains over about 80% of the surface, and a uniform crater population that implies a mean surface age of only 300 to 700 Myr.



Nili Fossae, Mars: THEMIS thermal inertia (left) and CRISM mineralogy (right) with Mg-carbonates (green), olivine sands (yellow) and basalt (blue); scale bars 20 km. Credit: NASA/JPL-Caltech/ASU/JHU APL.

Reflectance spectroscopy reads minerals from their absorption bands (CRISM at Mars: clays, carbonates); we land at a few sites, and almost all surface knowledge comes from orbit.



Lunar crustal thickness from GRAIL gravity and topography, nearside (left) and farside (right).
Credit: NASA/MIT/GSFC/JPL-Caltech.

Gravity field mapping reveals buried mass: the crust is 34–43 km on average and thin under the big basins, so a basin is a hole through most of the crust; **laser altimetry** (MOLA) supplies the topography to about 1 m.



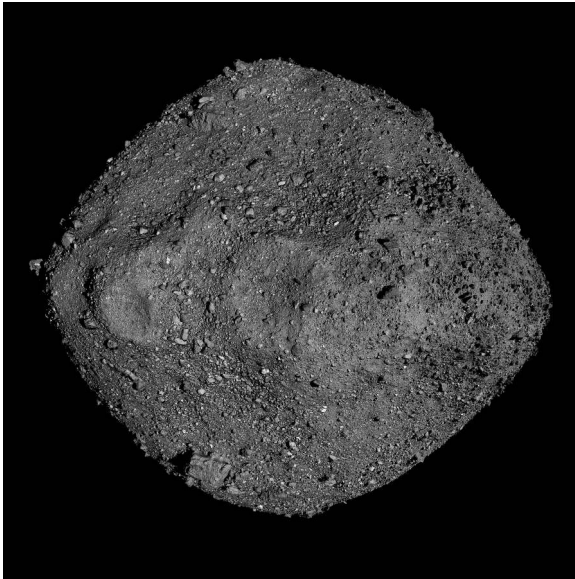
9. Regolith and space weathering

Apollo 11 bootprint in the lunar regolith. Credit: NASA



Apollo 11 bootprint in the lunar regolith. Credit: NASA/Apollo 11.

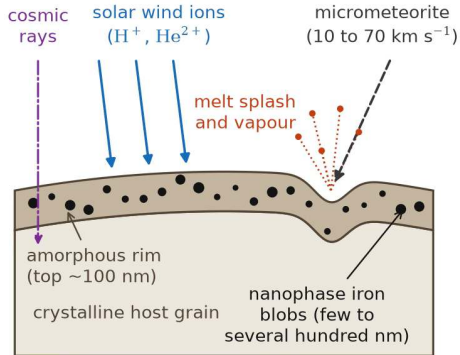
Regolith: loose debris made by impact gardening, 2–4 m on the maria and 6–8 m on the highlands; **space weathering** by solar wind and micrometeorites makes nanophase iron, so surfaces turn **darker and redder** and fresh craters like Tycho look bright.



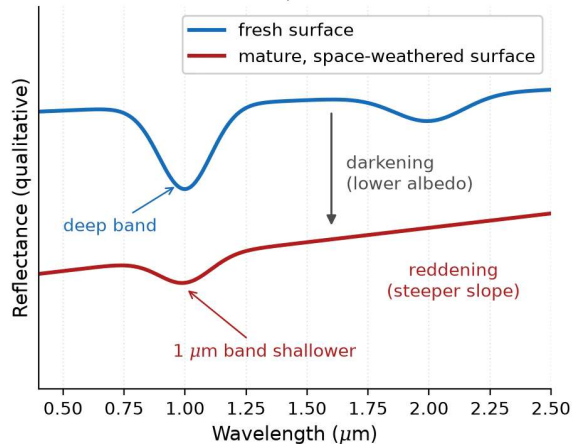
Bennu from OSIRIS-REx in December 2018, from 24 km: boulders at every scale on a body about 490 m across. Credit: NASA/Goddard/University of Arizona, public domain.

No smooth ponded plains and a very low bulk density, so high porosity: a rubble pile of collisional fragments reaccumulated under self-gravity; OSIRIS-REx returned 121.6 g of it to Earth in September 2023.

(a) Regolith grain cross-section (schematic)

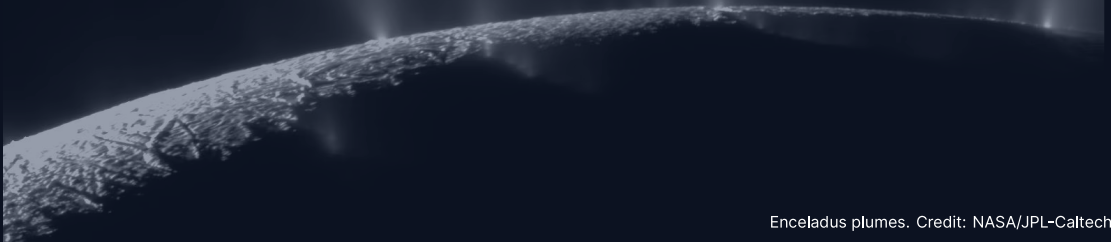


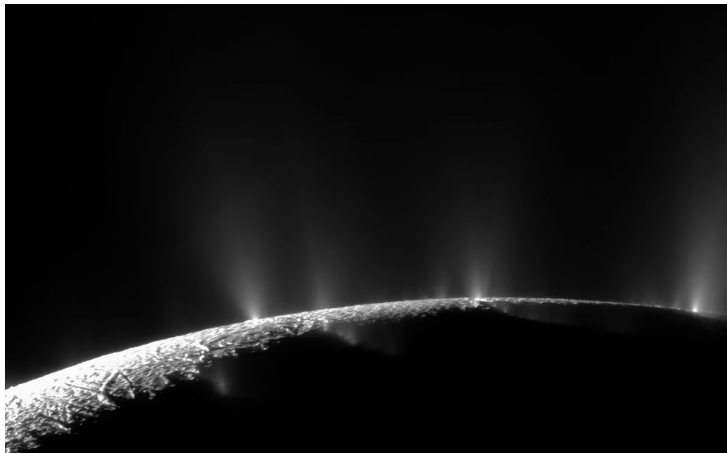
(b) Reflectance spectra (schematic)



Space weathering: solar wind ions and micrometeorites build an amorphous rim with nanophase iron; mature surfaces are darker, redder and show a shallower $1\text{ }\mu\text{m}$ band. Course figure.

10. Cryovolcanism on icy bodies



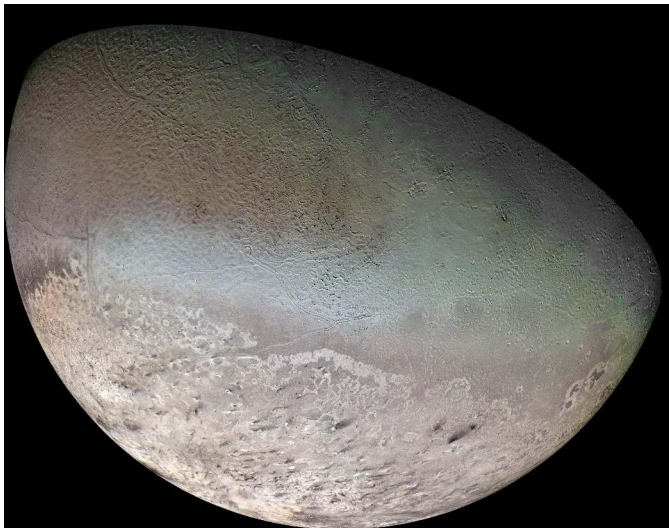


Water plumes from the south pole of Enceladus, Cassini. Credit: NASA/JPL-Caltech/SSI.

The “magma” is water, ammonia-water or methane at 100–270 K (nitrogen gas at about 38 K on Triton); **tidal heating** keeps oceans liquid far outside the habitable zone: Enceladus erupts, Europa resurfaces, Triton vents (Lecture 11).



Double ridges and lineae near Pwyll crater on Europa, Galileo SSI: so few craters that the surface is only 40–90 Myr old. Credit: NASA/JPL-Caltech/University of Arizona/University of Colorado.



Voyager 2 mosaic of Triton: nitrogen geyser streaks cross the pink N_2/CH_4 polar frost (bottom), cantaloupe terrain of dimples and ridges fills the north. Credit: NASA/JPL/Voyager

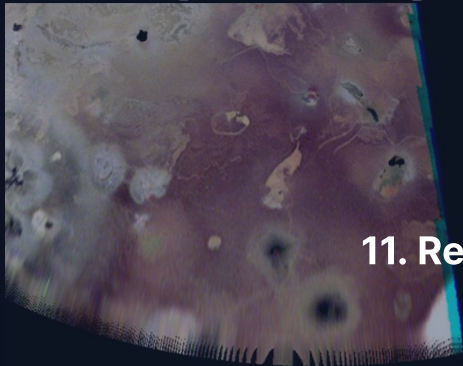
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Pluto in enhanced true colour from New Horizons, 14 July 2015: Sputnik Planitia is the bright western lobe of the heart-shaped Tombaugh Regio. Credit: NASA/JHUAPL/SwRI, public domain.

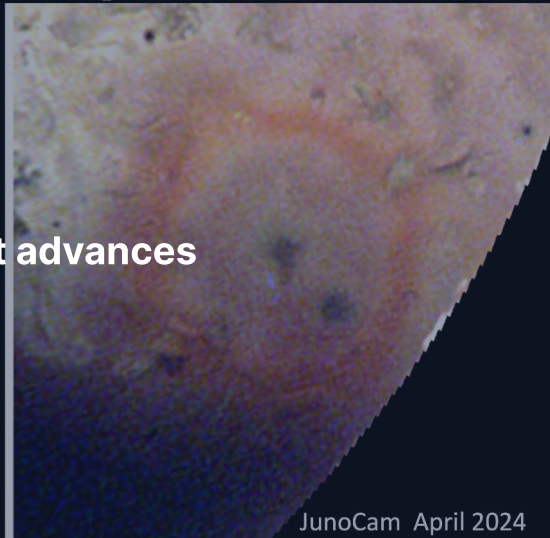
A crater-free plain of convecting nitrogen ice bordered by water-ice mountains, and dark reddish terrains coated in tholins from radiation chemistry: a small, distant dwarf planet can be geologically active today.

Change: New Bright Red Ring at Nusku



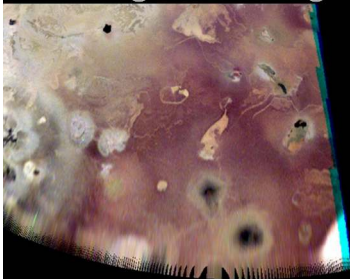
JunoCam February 2024

11. Recent advances



JunoCam April 2024

Change: New Bright Red Ring at Nusku



JunoCam February 2024



JunoCam April 2024

The Nusku region on Io in February and April 2024, JunoCam: a red ring of fresh sulfur-rich pyroclastic deposits appears between two flybys, resurfacing visible within months. Credit: NASA/JPL-Caltech/SwRI/MSSS/Jason Perry.

Summary

1. Surfaces record **endogenic** (volcanism, tectonics) against **exogenic** (impacts, erosion) processes; crater counting plus Apollo calibration gives absolute ages
2. Crater scaling law: $D \sim (E/\rho g)^{1/4}$, a fourth root of the energy
3. No plate tectonics: giant volcanoes, one rift, a shrinking Mercury; Earth alone breaks its lid, and fluvial features on Mars are strong evidence for past liquid water

Next: Lecture 8 goes underneath, to the interiors that drive the endogenic half of this record.

Questions?

Next: Lecture 8, Planetary interiors: structure, composition & dynamics

Thu 1 Oct, 11:00–13:00, room 5419.0161 (Collegezaal)

Next tutorial: Fri 2 Oct, 13:00–15:00, room 5419.0161 (Collegezaal)

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